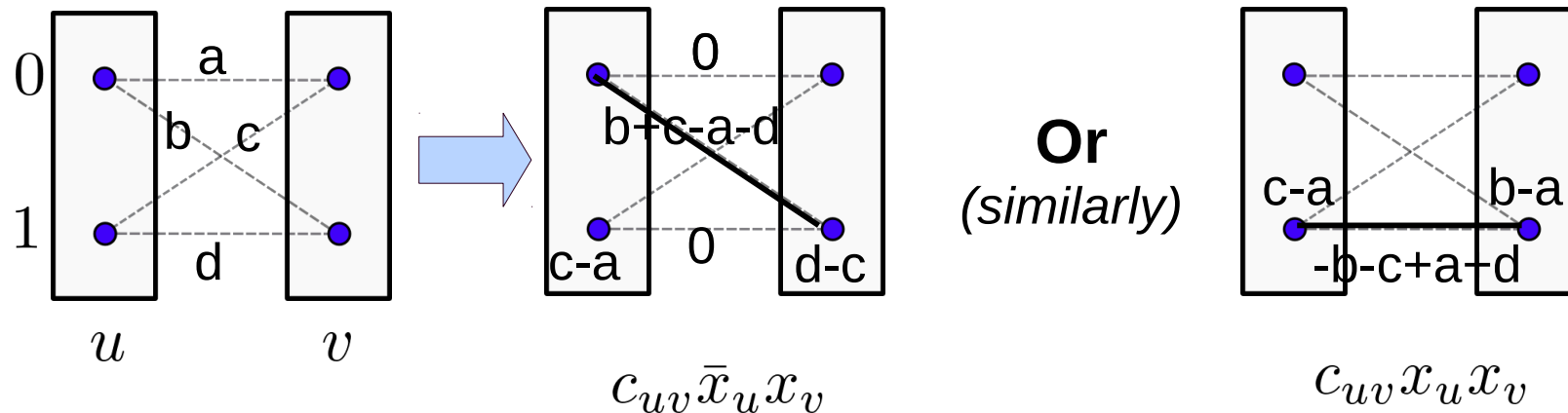


Persistency and Fusion Moves

Binary LP Relaxation as Min-Cut

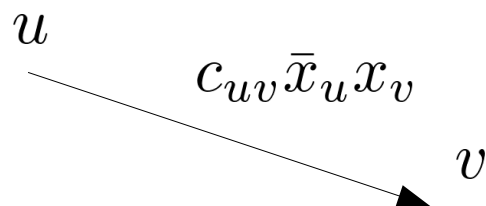
1. Reparametrize energy and write it as QPBF



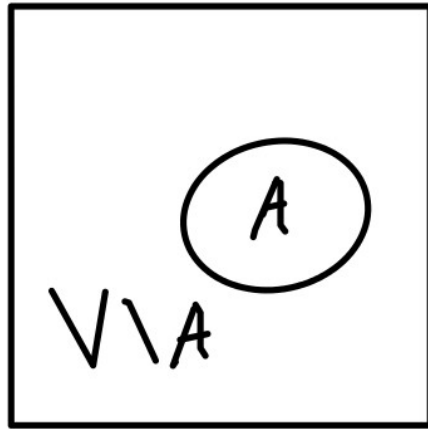
2. Introduce additional variables to obtain an oriented QPBF

$$c_{uv}x_u x_v \rightarrow \frac{1}{2}c_{uv}x_u \bar{y}_v + \frac{1}{2}c_{uv}\bar{y}_u x_v \quad \text{s.t. } x_u = \bar{y}_u, \quad u \in \mathcal{V}$$

3. Turn O-QPBF to Min-Cut and solve it. Reconstruct a relaxed labeling



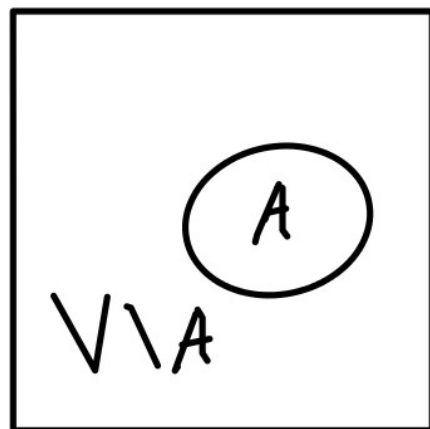
$$\mu'_v(1) = \begin{cases} x_v, & x_v = \bar{y}_v \\ \frac{1}{2}, & x_v \neq \bar{y}_v \end{cases}$$



A-integer Labels
V/A-fractional Labels

Definition:

$x^{\mathcal{A}}$ – (weakly) persistent, if $x^{\mathcal{A}} = x^*|_{\mathcal{A}}$, where $x^* \in \arg \min_{x \in \mathcal{X}} \langle \theta, \delta(x) \rangle$



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Persistency criterion:

$x^{\mathcal{A}}$ – (weakly) persistent, if for all $\tilde{x} \in \mathcal{X}_{V \setminus \mathcal{A}}$ it holds:

$$x^{\mathcal{A}} \in \arg \min_{x \in \mathcal{X}_{\mathcal{A}}} E((x, \tilde{x}))$$

Theorem: Integer part of any solution to the LP relaxation of binary energy is persistent

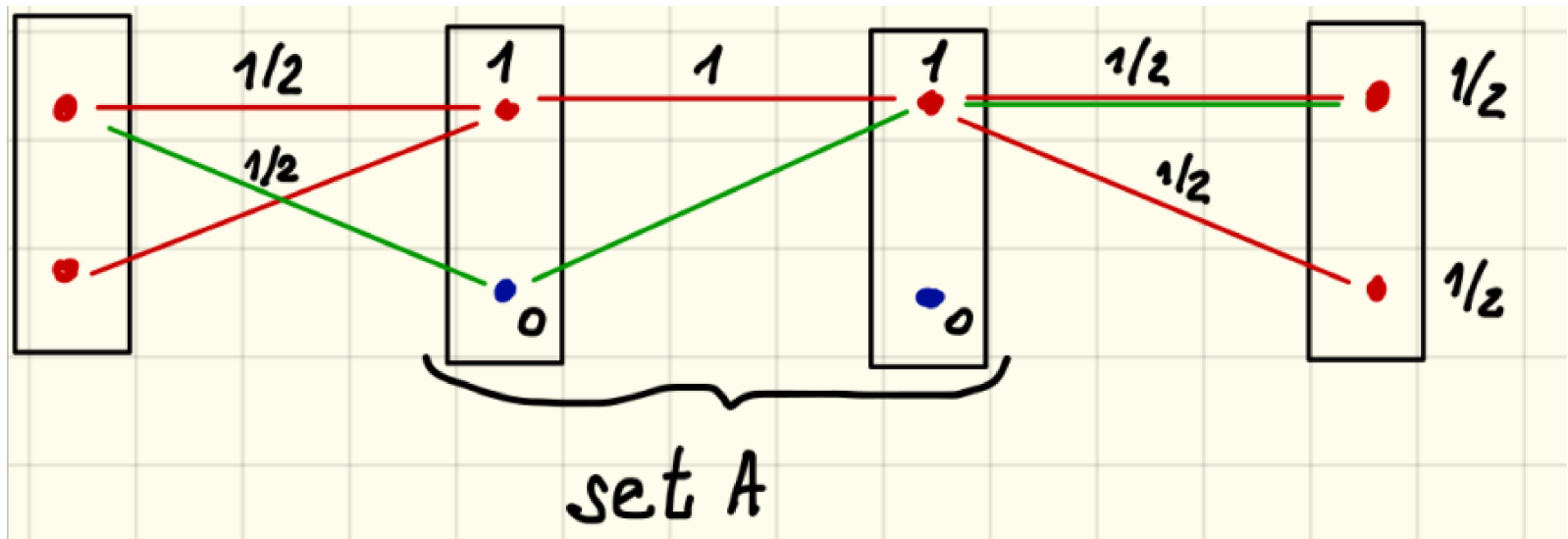
$\mathcal{A}(\mu) = \{v \in \mathcal{V} : \mu_v \in \{0, 1\}^2\}$ - set of integer coordinates

Let $\mu' \in \arg \min_{\mu \in \mathcal{L}} \langle \theta, \mu \rangle$. Then $\mu'|_{\mathcal{A}(\mu')}$ is persistent

Theorem: Integer part of any solution to the LP relaxation of binary energy is persistent

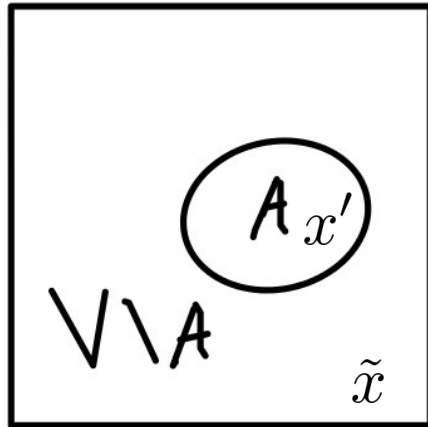
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Does the persistency property holds in the multi-label case?

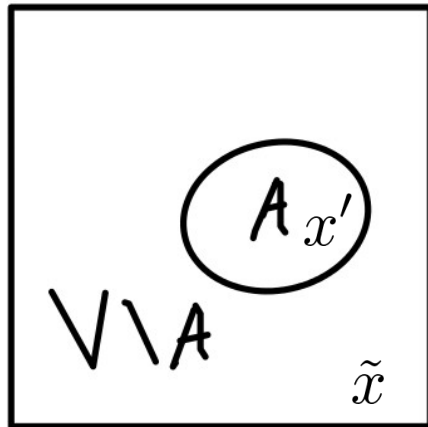
- 1) yes, there is no difference
- 2) no, because the proof works only in the binary case
- 3) no, because we had a counterexample earlier
(partial integrality does not imply partial optimality)
- 4) no, because I know a counterexample and can give it
- 5) 2)-3) are both correct
- 6) 2)-3) are both correct and 4) holds for me



A-integer Labels
V/A-fractional Labels

$$x' = \mu' |_{\mathcal{A}(\mu')}$$

$$E((x, \tilde{x})) \geq E((x', \tilde{x})), \quad \forall x \in \mathcal{X}_{\mathcal{A}} \text{ and } \tilde{x} \in \mathcal{X}_{V \setminus \mathcal{A}}$$



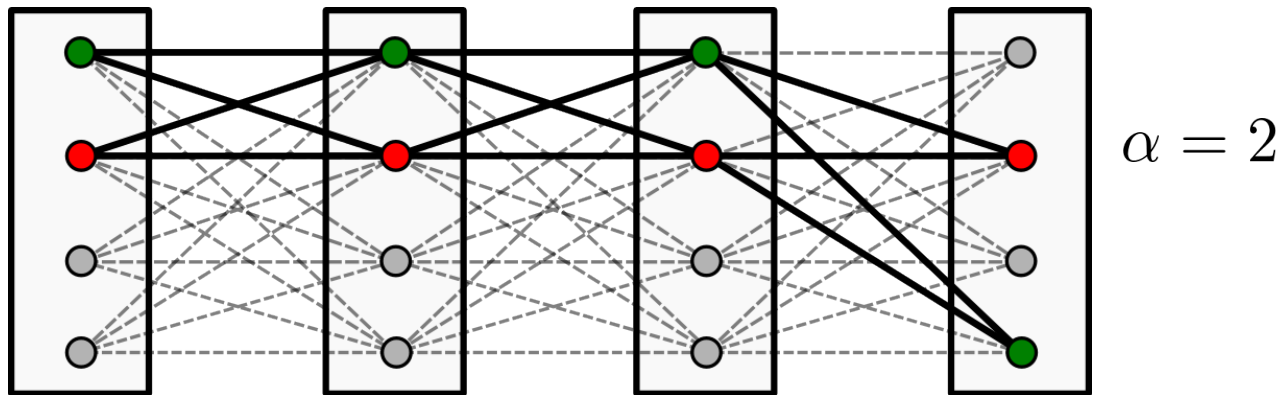
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α -Fusion moves:

Select α to expand, solve the binary problem

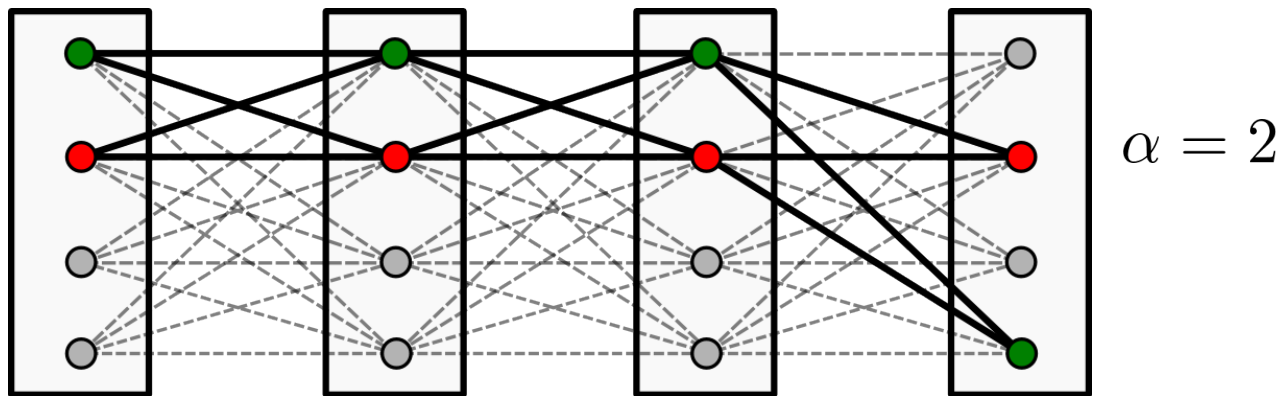


Properties:

- +1) monotonous (energy non-increasing)
- +2) applicable to any potentials
- +3) other move strategies can be used
- 4) twice as big min-cut subproblems, as in alpha-expansion
- 5) no optimality guarantees
- 6) no optimality bounds

α -Fusion moves:

Select α to expand, solve the binary problem



LP relaxation of binary energies:

1. Arc-consistency = relaxed optimality (can be solved by TRW-S)
2. Can be solved by min-cut
3. Has a persistency property
4. Can be used for monotonous fusion move algorithms